Lesson 04 material: The Research Cycle and Finding a Research Topic

2.1: Prelude to the Scientific Method

From *Research Methods in Psychology (4th American edition)* by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0). This section: CC BY-NC-SA 4.0; authored, remixed or curated by LibreTexts.
Original page, with audio and online exercises:
https://socialsci.libretexts.org/Bookshelves/Psychology/Research_Methods_and_Statistics/Research_Methods_in_Psychology_(Jhangiani_Chiang_Cuttler_and_Leighton)/02
Changes: formatted for print; exercise answers are in the study guide.

Here is the abstract of a 2014 article in the journal *Psychological Science*.

Taking notes on laptops rather than in longhand is increasingly common. Many researchers have suggested that laptop note taking is less effective than longhand note taking for learning. Prior studies have primarily focused on students' capacity for multitasking and distraction when using laptops. The present research suggests that even when laptops are used solely to take notes, they may still be impairing learning because their use results in shallower processing. In three studies, we found that students who took notes on laptops performed worse on conceptual questions than students who took notes longhand. We show that whereas taking more notes can be beneficial, laptop note takers' tendency to transcribe lectures verbatim rather than processing information and reframing it in their own words is detrimental to learning. (Mueler & Oppenheimer, 2014, p. 1159)[1]

In this abstract, the researcher has identified a research question—about the effect of taking notes on a laptop on learning—and identified why it is worthy of investigation—because the practice is ubiquitous and may be harmful to learning.

References

Mueller, P. A., & Oppenheimer, D. M. (2014). The pen is mightier than the keyboard: Advantages of longhand over laptop note taking. *Psychological Science, 25*(6), 1159-1168. ↵

2.1: Prelude to the Scientific Method is shared under a CC BY-NC-SA 4.0 license and was authored, remixed, and/or curated by LibreTexts.

2.2: A Model of Scientific Research in Psychology

From *Research Methods in Psychology (4th American edition)* by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0). This section: CC BY-NC-SA 4.0; authored, remixed or curated by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton via source content that was edited to the style and standards of the LibreTexts platform.
Original page, with audio and online exercises:
https://socialsci.libretexts.org/Bookshelves/Psychology/Research_Methods_and_Statistics/Research_Methods_in_Psychology_(Jhangiani_Chiang_Cuttler_and_Leighton)/02
Changes: formatted for print; exercise answers are in the study guide.

Learning Objectives

1. Review a general model of scientific research in psychology.

Figure presents a simple model of scientific research in psychology. The researchers formulate a research question, conduct an empirical study designed to answer the question, analyze the resulting data, draw conclusions about the answer to the question, and publishes the results so that they become part of the research literature (i.e., all the published research in that field). Because the research literature is one of the primary sources of new research questions, this process can be thought of as a cycle. New research leads to new questions, which lead to new research, and so on. Figure also indicates that research questions can originate outside of this cycle either with informal observations or with practical problems that need to be solved. But even in these cases, the researcher would start by checking the research literature to see if the question had already been answered and to refine it based on what previous research had already found.

Figure: A Simple Model of Scientific Research in Psychology

The research by Mehl and his colleagues is described nicely by this model. Their research question—whether women are more talkative than men—was suggested to them both by people's stereotypes and by claims published in the research literature about the relative talkativeness of women and men. When they checked the research literature, however, they found that this question had not been adequately addressed in scientific studies. They then conducted a careful empirical study, analyzed the results (finding very little difference between women and men), formed their conclusions, and published their work so that it became part of the research literature. The publication of their article is not the end of the story, however, because their work suggests many new questions (about the reliability of the result, about potential cultural differences, etc.) that will likely be taken up by them and by other researchers inspired by their work.

Video: https://youtu.be/XToWVxS_9lA

Cognitive neuroscientist David Strayer, PhD, studies how the brain processes information to better understand what causes driver distraction. By determining what takes our attention off the road, he's helping to develop regulatory policies that keep us safer behind the wheel.
https://youtu.be/XToWVxS_9lA

As another example, consider that as cell phones became more widespread during the 1990s, people began to wonder whether, and to what extent, cell phone use had a negative effect on driving. Many psychologists decided to tackle this question scientifically (e.g., Collet, Guillot, & Petit, 2010)[1]. It was clear from previously published research that engaging in a simple verbal task impairs performance on a perceptual or motor task carried out at the same time, but no one had studied the effect specifically of cell phone use on driving. Under carefully controlled conditions, these researchers compared people's driving performance while using a cell phone with their performance while not using a cell phone, both in the lab and on the road. They found that people's ability to detect road hazards, reaction time, and maintain control of the vehicle were all impaired by cell phone use. Each new study was published and became part of the growing research literature on this topic. For instance, other research teams subsequently demonstrated that cell phone conversations carry a greater risk than conversations with a passenger who is aware of driving conditions, which often become a point of conversation (e.g., Drews, Pasupathi, & Strayer, 2004)[2].

References

1. Collet, C., Guillot, A., & Petit, C. (2010). Phoning while driving I: A review of epidemiological, psychological, behavioral and physiological studies. *Ergonomics, 53*, 589–601. ↵
2. Drews, F. A., Pasupathi, M., & Strayer, D. L. (2004). Passenger and cell-phone conversations in simulated driving. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 48*, 2210–2212. ↵

Links on the original page

- Video (YouTube): https://youtu.be/XToWVxS_9lA

2.3: Finding a Research Topic

From *Research Methods in Psychology (4th American edition)* by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0). This section: CC BY-NC-SA 4.0; authored, remixed or curated by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton via source content that was edited to the style and standards of the LibreTexts platform.
Original page, with audio and online exercises:
https://socialsci.libretexts.org/Bookshelves/Psychology/Research_Methods_and_Statistics/Research_Methods_in_Psychology_(Jhangiani_Chiang_Cuttler_and_Leighton)/02
Changes: formatted for print; exercise answers are in the study guide.

Learning Objectives

1. Learn some common sources of research ideas.
2. Define the research literature in psychology and give examples of sources that are part of the research literature and sources that are not.
3. Describe and use several methods for finding previous research on a particular research idea or question.

Good research must begin with a good research question. Yet coming up with good research questions is something that novice researchers often find difficult and stressful. One reason is that this is a creative process that can appear mysterious—even magical—with experienced researchers seeming to pull interesting research questions out of thin air. However, psychological research on creativity has shown that it is neither as mysterious nor as magical as it appears. It is largely the product of ordinary thinking strategies and persistence (Weisberg, 1993)[1]. This section covers some fairly simple strategies for finding general research ideas, turning those ideas into empirically testable research questions, and finally evaluating those questions in terms of how interesting they are and how feasible they would be to answer.

Finding Inspiration

Research questions often begin as more general research ideas—usually focusing on some behavior or psychological characteristic: talkativeness, learning, depression, bungee jumping, and so on. Before looking at how to turn such ideas into empirically testable research questions, it is worth looking at where such ideas come from in the first place. Three of the most common sources of inspiration are informal observations, practical problems, and previous research.

Informal observations include direct observations of our own and others' behavior as well as secondhand observations from non-scientific sources such as newspapers, books, blogs, and so on. For example, you might notice that you always seem to be in the slowest moving line at the grocery store. Could it be that most people think the same thing? Or you might read in a local newspaper about people donating money and food to a local family whose house has burned down and begin to wonder about who makes such donations and why. Some of the most famous research in psychology has been inspired by informal observations. Stanley Milgram's famous research on obedience to authority, for example, was inspired in part by journalistic reports of the trials of accused Nazi war criminals—many of whom claimed that they were only obeying orders. This led him to wonder about the extent to which ordinary people will commit immoral acts simply because they are ordered to do so by an authority figure (Milgram, 1963)[2].

Practical problems can also inspire research ideas, leading directly to applied research in such domains as law, health, education, and sports. Does taking lecture notes by hand improve students' exam performance? How effective is psychotherapy for depression compared to drug therapy? To what extent do cell phones impair people's driving ability? How can we teach children to read more efficiently? What is the best mental preparation for running a marathon?

Probably the most common inspiration for new research ideas, however, is previous research. Recall that science is a kind of large-scale collaboration in which many different researchers read and evaluate each other's work and conduct new studies to build on it. Of course, experienced researchers are familiar with previous research in their area of expertise and probably have a long list of ideas. This suggests that novice researchers can find inspiration by consulting with a more experienced researcher (e.g., students can consult a faculty member). But they can also find inspiration by picking up a copy of almost any professional journal and reading the titles and abstracts. In one typical issue of *Psychological Science*, for example, you can find articles on the perception of shapes, anti-Semitism, police lineups, the meaning of death, second-language learning, people who seek negative emotional experiences, and many other topics. If you can narrow your interests down to a particular topic (e.g., memory) or domain (e.g., health care), you can also look through more specific journals, such as *Memory & Cognition* or *Health Psychology*.

Video: https://youtu.be/nXNztCLYgxc

In this video, you will learn how to develop a good research topic: https://www.youtube.com/embed/nXNztCLYgxc.

Research literature in any field is all the published research in that field. Reviewing the research literature means finding, reading, and summarizing the published research relevant to your topic of interest. In addition to helping you discover new research questions, reviewing the literature early in the research process can help you in several other ways.

- It can tell you if a research question has already been answered.
- It can help you evaluate the interestingness of a research question.
- It can give you ideas for how to conduct your own study.
- It can tell you how your study fits into the research literature.

The research literature in psychology is enormous—including millions of scholarly articles and books dating to the beginning of the field—and it continues to grow. Although its boundaries are somewhat fuzzy, the research literature definitely does not include self-help and other pop psychology books, dictionary and encyclopedia entries, websites, and similar sources that are intended mainly for the general public. These are considered unreliable because they are not reviewed by other researchers and are often based on little more than common sense or personal experience. Wikipedia contains much valuable information, but because its authors are anonymous and may not have any formal training or expertise in that subject area, and its content continually changes it is unsuitable as a basis of sound scientific research. For our purposes, it helps to define the research literature as consisting almost entirely of two types of sources: articles in professional journals, and scholarly books in psychology and related fields.

Professional Journals

Professional journals are periodicals that publish original research articles. There are thousands of professional journals that publish research in psychology and related fields. They are usually published monthly or quarterly in individual issues, each of which contains several articles. The issues are organized into volumes, which usually consist of all the issues for a calendar year. Some journals are published in hard copy only, others in both hard copy and electronic form, and still others in electronic form only.

Most articles in professional journals are one of two basic types: empirical research reports and review articles. **Empirical research reports** describe one or more new empirical studies conducted by the authors. They introduce a research question, explain why it is interesting, review previous research, describe their method and results, and draw their conclusions. **Review articles** summarize previously published research on a topic and usually present new ways to organize or explain the results. When a review article is devoted primarily to presenting a new theory, it is often referred to as a **theoretical article**. When a review article provides a statistical summary of all of the previous results it is referred to as a **meta-analysis**.

Figure: Small Sample of the Thousands of Professional Journals That Publish Research in Psychology and Related Fields

Most professional journals in psychology undergo a process of **double-blind peer review**. Researchers who want to publish their work in the journal submit a manuscript to the editor—who is generally an established researcher too—who in turn sends it to two or three experts on the topic. Each reviewer reads the manuscript, writes a critical but constructive review, and sends the review back to the editor along with recommendations about whether the manuscript should be published or not. The editor then decides whether to accept the article for publication, ask the authors to make changes and resubmit it for further consideration, or reject it outright. In any case, the editor forwards the reviewers' written comments to the researchers so that they can revise their manuscript accordingly. This entire process is double-blind, as the reviewers do not know the identity of the researcher(s) and vice versa. Double-blind peer review is helpful because it ensures that the work meets basic standards of the field before it can enter the research literature. However, in order to increase transparency and accountability, some newer open access journals (e.g., Frontiers in Psychology) utilize an open peer review process wherein the identities of the reviewers (which remain concealed during the peer review process) are published alongside the journal article.

Scholarly Books

Scholarly books are books written by researchers and practitioners mainly for use by other researchers and practitioners. A **monograph** is written by a single author or a small group of authors and usually, gives a coherent presentation of a topic much like an extended review article. **Edited volumes** have an editor or a small group of editors who recruit many authors to write separate chapters on different aspects of the same topic. Although edited volumes can also give a coherent presentation of the topic, it is not unusual for each chapter to take a different perspective or even for the authors of different chapters to openly disagree with each other. In general, scholarly books undergo a peer review process similar to that used by professional journals.

Literature Search Strategies

Using PsycINFO and Other Databases

The primary method used to search the research literature involves using one or more electronic databases. These include Academic Search Premier, JSTOR, and ProQuest for all academic disciplines, ERIC for education, and PubMed for medicine and related fields. The most important for our purposes, however, is PsycINFO, which is produced by the American Psychological Association (APA). **PsycINFO** is so comprehensive—covering thousands of professional journals and scholarly books going back more than 100 years—that for most purposes its content is synonymous with the research literature in psychology. Like most such databases, PsycINFO is usually available through your university library.

PsycINFO consists of individual records for each article, book chapter, or book in the database. Each record includes basic publication information, an abstract or summary of the work (like the one presented at the start of this chapter), and a list of other works cited by that work. A computer interface allows entering one or more search terms and returns any records that contain those search terms. (These interfaces are provided by different vendors and therefore can look somewhat different depending on the library you use.) Each record also contains lists of keywords that describe the content of the work and also a list of index terms. The index terms are especially helpful because they are standardized. Research on differences between females and males, for example, is always indexed under "Human Sex Differences." Research on note-taking is always indexed under the term "Learning Strategies." If you do not know the appropriate index terms, PsycINFO includes a thesaurus that can help you find them.

Given that there are nearly four million records in PsycINFO, you may have to try a variety of search terms in different combinations and at different levels of specificity before you find what you are looking for. Imagine, for example, that you are interested in the question of whether males and females differ in terms of their ability to recall experiences from when they were very young. If you were to enter the search term "memory," it would return far too many records to look through individually. This is where the thesaurus helps. Entering "memory" into the thesaurus provides several more specific index terms—one of which is "early memories." While searching for "early memories" among the index terms still returns too many to look through individually—combining it with "human sex differences" as a second search term returns fewer articles, many of which are highly relevant to the topic.

Depending on the vendor that provides the interface to PsycINFO, you may be able to save, print, or e-mail the relevant PsycINFO records. The records might even contain links to full-text copies of the works themselves. (PsycARTICLES is a database that provides full-text access to articles in all journals published by the APA.) If not, and you want a copy of the work, you will have to find out if your library carries the journal or has the book and the hard copy on the library shelves. Be sure to ask a librarian if you need help.

Video: https://youtu.be/fhhctbaVXvk

This is a sample search of PsycINFO on the EBSCOhost platform. We use a search example question to illustrate how you can use the elements of a question to build a search with the thesaurus. https://youtu.be/fhhctbaVXvk

Using Other Search Techniques

In addition to entering search terms into PsycINFO and other databases, there are several other techniques you can use to search the research literature. First, if you have one good article or book chapter on your topic—a recent review article is best—you can look through the reference list of that article for other relevant articles, books, and book chapters. In fact, you should do this with any relevant article or book chapter you find. You can also start with a

classic article or book chapter on your topic, find its record in PsycINFO (by entering the author's name or article's title as a search term), and link from there to a list of other works in PsycINFO that cite that classic article. This works because other researchers working on your topic are likely to be aware of the classic article and cite it in their own work. You can also do a general Internet search using search terms related to your topic or the name of a researcher who conducts research on your topic. This might lead you directly to works that are part of the research literature (e.g., articles in open-access journals or posted on researchers' own websites). The search engine Google Scholar is especially useful for this purpose. A general Internet search might also lead you to websites that are not part of the research literature but might provide references to works that are. Finally, you can talk to people (e.g., your instructor or other faculty members in psychology) who know something about your topic and can suggest relevant articles and book chapters.

Video: https://youtu.be/t1ZwgDeX2eQ

This tutorial explains how to use advanced features of Google Scholar. https://youtu.be/t1ZwgDeX2eQ

What to Search For

When you do a literature review, you need to be selective. Not every article, book chapter, and book that relates to your research idea or question will be worth obtaining, reading, and integrating into your review. Instead, you want to focus on sources that help you do four basic things: (a) refine your research question, (b) identify appropriate research methods, (c) place your research in the context of previous research, and (d) write an effective research report. Several basic principles can help you find the most useful sources.

First, it is best to focus on recent research, keeping in mind that what counts as recent depends on the topic. For newer topics that are actively being studied, "recent" might mean published in the past year or two. For older topics that are receiving less attention right now, "recent" might mean within the past 10 years. You will get a feel for what counts as recent for your topic when you start your literature search. A good general rule, however, is to start with sources published in the past five years. The main exception to this rule would be classic articles that turn up in the reference list of nearly every other source. If other researchers think that this work is important, even though it is old, then, by all means, you should include it in your review.

Second, you should look for review articles on your topic because they will provide a useful overview of it—often discussing important definitions, results, theories, trends, and controversies—giving you a good sense of where your own research fits into the literature. You should also look for empirical research reports addressing your question or similar questions, which can give you ideas about how to measure your variables and collect your data. As a general rule, it is good to use methods that others have already used successfully unless you have good reasons not to. Finally, you should look for sources that provide information that can help you argue for the interestingness of your research question. For a study on the effects of cell phone use on driving ability, for example, you might look for information about how widespread cell phone use is, how frequent and costly motor vehicle crashes are, and so on.

How many sources are enough for your literature review? This is a difficult question because it depends on how extensively your topic has been studied and also on your own goals. One study found that across a variety of professional journals in psychology, the average number of sources cited per article was about 50 (Adair & Vohra, 2003)[3]. This gives a rough idea of what professional researchers consider to be adequate. As a student, you might be assigned a much lower minimum number of references to include, but the principles for selecting the most useful ones remain the same.

References

1. Weisberg, R. W. (1993). *Creativity: Beyond the myth of genius*. New York, NY: Freeman. ↲
2. Milgram, S. (1963). Behavioral study of obedience. *Journal of Abnormal and Social Psychology, 67*, 371–378. ↲
3. Adair, J. G., & Vohra, N. (2003). The explosion of knowledge, references, and citations: Psychology's unique response to a crisis. *American Psychologist, 58*, 15–23. ↲

Links on the original page

- Video (YouTube): https://youtu.be/nXNztCLYgxc
- Frontiers in Psychology: https://www.frontiersin.org/journals/psychology#
- Video (YouTube): https://youtu.be/fhhctbaVXvk
- Video (YouTube): https://youtu.be/t1ZwgDeX2eQ